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CIA - 1

course name : Computer Networks
course code : CSA0710

① a) Explain the Data link and Transport layers.

Datalink layer:

The data link layer provides communication b/w directly connection devices its main function are.

framing: Divides data into frames
MAC Addressing: identifies devices on a local network.
Error Detection: Detect errors using CRC.
Flow Control: Controls the speed of data transmission
Access Control: Manages access to the communication medium.

Transport layer:

The Transport layer provides communication b/w application on different devices.

Segmentation: Divides data into smaller segments.
Port Addressing: Identifies applications.
Error Recovery: Prevents receiver overload.
Reliable Delivery: TCP provides overloaded and reliable data delivery.

b)

Connection-oriented	Connectionless
> Establishes a Connection first > TCP provides reliable delivery Ex: file transfer using TCP	> Does not require connection setup > UDP does not guarantee delivery Ex: DNS queries using UDP.

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(2) a) Encapsulation and Decapsulation:

Encapsulation: Each OSI layer adds it header as data moves from sender to receiver's network.

Decapsulation: the receiver removes headers layer by layer to obtain the original data.

Order: Data -> Segment -> Packet -> Frame -> Bits.

b) Network Layer:

The network layer performs routing b/w networks. It uses logical IP addresses to identify destination and forwards packet through routers.

a) Classful Addressing and CIDR

classful addressing uses fixed network size. it wastes IP addresses and provides limit flexibility. CIDR user flexible Prefix lengths such as 126. it reduces address wastage and supports route aggregation.

b) subnet calculation:

Given IP: 192.168.25.130
mask : 255.255.255.192 (126)
network ID: 192.168.25.128.
Broadcast : 192.168.25.191
first usable : 192.168.25.129.
Last usable : 192.168.25.190
usable hosts : 62

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(4) a) subnet mask:

Given: 172.16.0.0/16 & at least 6 subnets

$2^3 = 8$ subnets

Brrowed bits = 3
new CIDR : /19
Subnet mask : 255.255.224.0

b) Subnets and Hosts

Total subnets: $2^3 = 8$

Host bits: $32 - 19 = 13$

Usable hosts: $2^{13} - 2 = 8190$

c) First two subnets:

Details	Subnet 1	Subnet 2
Network ID	172.16.0.0	172.16.32.0
First Usable	172.16.0.1	172.16.32.1
Last Usable	172.16.31.254	172.16.63.254
Broad Cast	172.16.31.255	172.16.63.255

(5) a) Static vs Dynamic Routing:

Static routing: Routes are configured manually suitable for small, stable networks.

Dynamic routing: Routes are learned automatically using protocols such as RIP and OSPF.

-) Static routing uses fewer resources while dynamic routing adapts to network changes.

b) static route Configuration:

Router > enable
Router # Configure terminal
Router (Config) # ip route 192.168.2.0
~~Router C~~
255.255.255.0 10.0.0.2
Router (Config) # end
Router # show ip route: